

Smart-LMS Enterprise Engineering Manual

Complete Step-by-Step Architecture, Data Flows, Algorithms & Schemas

PRODUCTION GRADE • FULL-STACK AI LEARNING MANAGEMENT SYSTEM

1. Executive Summary & Core Topology

Smart-LMS is an enterprise-grade academic learning management system built with a decoupled Client-Server architecture. The frontend is a React 18 Single Page Application powered by Vite and Tailwind CSS v4. The backend is a Node.js & Express API monolith integrated with MongoDB, Socket.IO WebSockets, and multiple LLM providers (Groq LLaMA 3.1, Google Gemini, OpenAI).

2. Step-by-Step Engineering Specifications by Module

Module 1: AI Tutor & Academic Problem Solver (`/ai-tutor` , `/api/ai`)

Step-by-Step Execution Sequence

- Dispatch:** Client sends `POST /api/ai/solve` containing question, topic, and difficulty.
- Auth Guard:** `auth.middleware.js` decodes Bearer JWT token and validates user session.
- 30s Timeout Race:** `safeGroqCall()` executes `Promise.race()` between Groq LLaMA 3.1 completion and a 30-second rejection timer.
- Activity & Streak Mutation:** Updates user streak, appends today's `weeklyActivity` , and updates subject mastery formula:

$$\text{Mastery} = \min(100, \text{Questions} \times 5 + \text{Notes} \times 7 + \text{Quizzes} \times 10)$$

- Response:** Returns sanitized Markdown solution to the React frontend.

Module 2: Document Processing, Study Mode & OCR Engine (`/study` , `/upload`)

Step-by-Step Execution Sequence

- Binary Upload:** Multer interceptor sanitizes filename (`/[^a-zA-Z0-9]/g`), appends timestamp, and saves to `/uploads/` .
- OCR Extraction:** `Tesseract.recognize(filePath, 'eng')` processes uploaded images/PDFs using local training data (`eng.traineddata`).
- AI Flashcards/Notes Synthesis:** Extracted text is fed into Groq AI prompt templates to produce exam summaries and JSON flashcards.
- Persistence:** Stores structured notes inside MongoDB `Note` schema linked to user ID.

Module 3: AI Quiz & Mock Test Generator (`/quiz` , `/api/mocktest`)

Step-by-Step Execution Sequence

- MCQ Generation:** AI prompt enforces JSON output. Responses are parsed via `cleanAIResponse()` regex extractor.
- Client Exam Mode:** Client renders timed questionnaire with option selection.
- Evaluation & XP Grant:** Submits user answers to `POST /api/mocktest/submit` . Compares against answer key, calculates accuracy %, updates user total XP, level, and rank.

Module 4: Academic Planner & Analytics Engine (/planner , /analytics)

The analytics engine dynamically computes student risk scores, performance trends, and task impact reductions:

Risk Score Formula:

$\text{RiskScore} = \text{clamp}((100 - \text{Accuracy}) \times 0.4 + (100 - \text{Completion}) \times 0.3 + \text{StreakImpact} \times 0.3)$

where $\text{StreakImpact} = 80$ if $\text{Streak} \leq 2$, else $\max(0, 100 - \text{Streak} \times 5)$.

Risk Reduction on Task Completion:

Weak Practice Task: -6 • Mock Exam: -8 • Revision Task: -4 • Core Study: -3

Subject Confidence Score:

$\text{Confidence} = \text{clamp}(\text{Mastery} \times 0.6 + \text{LastMockAccuracy} \times 0.4)$

Module 5: Cognitive Brain AI & Gamification Engine (/critical , /api/brain)

Progression & Intelligence Scaling:

$\text{Level} = \lfloor \text{XP} / 100 \rfloor + 1$

$\text{IntelligenceScore} = 100 + \lfloor \text{XP} \times 0.8 \rfloor$

- **Ranks:** Beginner (<200 XP) → Intermediate (≥ 200) → Advanced (≥ 500) → Master (≥ 1000) → Grandmaster (≥ 2000).
- **Badges:** 🧠 Beginner Brain , 🧠 Smart Thinker , 🧠 Master Strategist , 🏆 Brain Legend , 🎯 Precision Mind (Accuracy $\geq 80\%$), 🔥 Focus Beast (Streak ≥ 10), ⚡ Streak Monster (Streak ≥ 25).

Module 6: Real-Time Social Engine, Group Study Rooms & WebRTC Signaling (/friends)

- **User Socket Registration:** `socket.on("register-user", userId)` stores mapping inside `onlineUsers` Map and joins `user_<id>` room.
- **Direct & Group Messaging:** Persists messages in MongoDB (`FriendMessage` / `GroupMessage`), calculates live online delivery status, and broadcasts to `group_<groupId>` .
- **WebRTC Video Calls:** Handshakes peer-to-peer audio/video connection via Socket.IO events (`call-user` , `incoming-call` , `answer-call` , `call-accepted`).

3. Complete Database Schema Directory

Model Name	File Path	Key Schema Fields	Indexes & Virtuals
User	src/models/user.model.js	name, username, email, password, exam, xp, level, rank, streak, accuracy, readiness, risk, dailyMission, weeklyActivity, subjectStats	Index: { xp: -1 } Virtual: levelProgress
Brain	src/models/brain.model.js	user, xp, level, intelligenceScore, gamesPlayed, correctAnswers, wrongAnswers, currentStreak, bestStreak, badges, dailyChallenge	Index: { intelligenceScore: -1 } Virtual: accuracy
Group	src/models/group.model.js	name, description, avatar, creator, members, admins, isPrivate	Index: { members: 1 }
GroupMessage	src/models/groupMessage.model.js	groupId, sender, content, messageType, attachmentType, fileUrl, fileSize, audioDuration, deliveredTo, seenBy	Index: { groupId: 1, createdAt: 1 }
FriendMessage	src/models/friendMessage.model.js	sender, recipient, content, isAttachment, attachmentType, fileUrl, isRead	Index: { sender: 1, recipient: 1 }
Planner	src/models/planner.model.js	user, weeklyPlan, riskScore, riskLevel, performanceTrend, weeklyCompletion, accuracyHistory	Index: { user: 1 }

4. Granular API Route Matrix

Route Path	Method	Protected	Payload / Parameters	Description
/api/auth/register	POST	No	{ name, email, password, exam }	Registers user, hashes password, returns JWT
/api/auth/login	POST	No	{ email, password }	Authenticates credentials, returns session JWT
/api/ai/solve	POST	Yes	{ question, topic, mode }	Invokes Groq LLaMA 3.1 LLM for step-by-step answer
/api/upload	POST	Yes	Multipart Binary File	Saves file to /uploads/ and returns static URL
/api/ocr/extract	POST	Yes	{ fileUrl }	Runs Tesseract OCR engine on document image
/api/mocktest/generate	POST	Yes	{ subject, questionCount }	Generates MCQ questionnaire JSON payload
/api/planner/my-plan	GET	Yes	None	Fetches adaptive planner tasks & risk metrics
/api/brain/submit-game	POST	Yes	{ isCorrect, xpEarned }	Records brain game outcome, updates badges & level
/api/groups/create	POST	Yes	{ name, memberIds }	Creates study group room & notifies members via Socket

5. Security & Reverse Proxy Rules

```
// 1. Trust proxy setting for Render Load Balancers
app.set("trust proxy", 1);

// 2. Helmet HTTP Security
app.use(helmet({ contentSecurityPolicy: false, crossOriginResourcePolicy: { policy: "cross-origin" } }));

// 3. Dynamic CORS Origin Validation
const isOriginAllowed = (origin) => {
  if (!origin || allowedOrigins.includes(origin) || /\.vercel\.app$/\.test(origin)) return true;
  return false;
};
```